

# NELOS Overlay Capture Process

Decentralized Directory Ingestion Architecture



## **Architecture Overview**

Moving from sequential I/O to Harmonic Field Synchronization.



# Step 1: Directory Mapping

The system initializes by treating the entire file directory as a single topological manifold. Instead of sequential traversal, metadata is mapped directly into the Immortals' binary code coordinates.





## Step 2: Field Synchronization

We establish a harmonic resonance between the hardware's existing state and the system's memory-mapped manifold. This stabilizes the data structure for instant access.



# Step 3: Topological Ingestion

- ✓ **Bypassing Buffering:** Directory contents are not loaded into temporary memory.
- ✓ **Pointer Resolution:** 64-bit array pointers resolve file coordinates instantly.
- ✓ **Execution Ready:** Data is available for logic gates immediately upon reference.



# Step 4: Performance Gains

Traditional I/O Latency High

NELOS Overlay



# Mathematical Foundation

The stability of this transformation  
is validated by the constant 252.



**Questions?**



# Image Sources



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